

## FOULA, United Kingdom

WMO: 030140

Lat: 60.117N

Lon: 2.067W

Elev: 13

StdP: 101.17

Time Zone: 0.00 (EUW)

Period: 94-16

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | -0.3       | 0.8 | -6.3                          | 2.2 | 1.5  | -5.1 | 2.5 | 2.1  | 22.6                  | 6.7  | 20.9 | 6.5  | 7.4                   | 340  | 0.863 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 2.2                    | 15.7            | 14.6 | 14.8 | 13.8 | 14.2 | 13.2 | 14.9                | 15.5 | 14.1 | 14.7 | 13.4 | 14.0 | 5.2                  | 90   |

| Dehumidification DP/MCDB and HR |      |      |      |     |      |      |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|-----|------|------|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |     |      | 2%   |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR  | MCDB | DP   | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e) | (f)  | (g)  | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 14.5                            | 10.3 | 15.3 | 13.7 | 9.8 | 14.5 | 13.0 | 9.3 | 13.8 | 41.7          | 15.6 | 39.6 | 14.7 | 37.8 | 14.1 | 25.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |     |
| (4) 19.4          | 17.4 | 15.3 | DB                         | -2.2 | 17.6               | 1.1 | 2.0                                                | -3.0 | 19.0       | -3.6 | 20.2       | -4.2 | 21.3       | -5.0 | 22.7 | (4) |
| (5)               |      |      | WB                         | -3.3 | 15.6               | 1.1 | 1.6                                                | -4.1 | 16.8       | -4.7 | 17.8       | -5.3 | 18.7       | -6.1 | 19.9 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 8.6    | 6.0   | 5.4   | 5.7   | 6.9   | 8.3   | 10.2  | 12.0  | 12.6  | 11.7  | 9.9   | 8.1   | 6.4   | (6)  |  |
|                                                                        | DBStd       | 3.14   | 1.98  | 2.28  | 2.00  | 1.72  | 1.57  | 1.32  | 1.37  | 1.21  | 1.57  | 1.99  | 2.47  | 2.57  | (7)  |  |
|                                                                        | HDD10.0     | 758    | 125   | 130   | 133   | 93    | 54    | 13    | 1     | 0     | 4     | 27    | 64    | 114   | (8)  |  |
|                                                                        | HDD18.3     | 3551   | 383   | 363   | 391   | 343   | 311   | 243   | 196   | 178   | 199   | 263   | 308   | 371   | (9)  |  |
|                                                                        | CDD10.0     | 249    | 0     | 0     | 0     | 0     | 2     | 19    | 63    | 81    | 54    | 22    | 6     | 2     | (10) |  |
|                                                                        | CDD18.3     | 1      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 0     | 0     | 0     | (11) |  |
|                                                                        | CDH23.3     | 3      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 3     | 0     | 0     | 0     | (12) |  |
|                                                                        | CDH26.7     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | (13) |  |
| (14) Wind                                                              | WSAvg       | 7.5    | 9.4   | 9.4   | 8.3   | 7.2   | 6.0   | 5.4   | 4.9   | 5.2   | 6.9   | 8.7   | 9.3   | 8.8   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 1360   | 173   | 135   | 129   | 74    | 60    | 65    | 71    | 93    | 111   | 136   | 148   | 165   | (15) |  |
|                                                                        | PrecMax     | 1622   | 272   | 237   | 205   | 117   | 107   | 117   | 129   | 138   | 163   | 195   | 228   | 262   | (16) |  |
|                                                                        | PrecMin     | 1080   | 79    | 71    | 50    | 29    | 17    | 17    | 33    | 49    | 36    | 55    | 62    | 70    | (17) |  |
|                                                                        | PrecStd     | 136    | 52    | 38    | 32    | 20    | 20    | 25    | 22    | 24    | 30    | 33    | 37    | 40    | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 10.1  | 9.7   | 9.3   | 10.6  | 13.1  | 15.7  | 17.7  | 17.2  | 16.8  | 13.9  | 12.8  | 13.1  | (19) |  |
|                                                                        |             | MCWB   | 9.0   | 8.4   | 8.1   | 9.0   | 11.0  | 13.5  | 16.5  | 15.8  | 15.0  | 12.4  | 11.5  | 12.2  | (20) |  |
|                                                                        | 2%          | DB     | 9.5   | 9.3   | 8.9   | 9.7   | 11.4  | 13.4  | 15.6  | 15.8  | 14.6  | 13.0  | 12.1  | 11.4  | (21) |  |
|                                                                        |             | MCWB   | 8.5   | 8.1   | 7.8   | 8.5   | 9.8   | 11.8  | 14.6  | 14.8  | 13.6  | 11.8  | 11.1  | 10.4  | (22) |  |
|                                                                        | 5%          | DB     | 9.0   | 8.8   | 8.5   | 9.2   | 10.7  | 12.5  | 14.5  | 14.9  | 14.0  | 12.5  | 11.6  | 10.3  | (23) |  |
|                                                                        |             | MCWB   | 8.0   | 7.7   | 7.5   | 8.2   | 9.3   | 11.2  | 13.6  | 14.0  | 13.1  | 11.3  | 10.6  | 9.3   | (24) |  |
|                                                                        | 10%         | DB     | 8.6   | 8.3   | 8.1   | 8.9   | 10.2  | 11.9  | 13.7  | 14.3  | 13.5  | 12.1  | 11.0  | 9.6   | (25) |  |
|                                                                        |             | MCWB   | 7.6   | 7.2   | 7.1   | 7.9   | 9.0   | 10.7  | 12.8  | 13.3  | 12.6  | 11.0  | 10.0  | 8.6   | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 9.5   | 9.1   | 8.7   | 9.5   | 11.4  | 13.7  | 16.8  | 16.1  | 15.2  | 12.9  | 12.1  | 12.3  | (27) |  |
|                                                                        |             | MCDB   | 9.7   | 9.3   | 8.9   | 10.0  | 12.6  | 15.3  | 17.5  | 16.8  | 15.9  | 13.3  | 12.4  | 13.0  | (28) |  |
|                                                                        | 2%          | WB     | 8.8   | 8.5   | 8.2   | 8.9   | 10.2  | 12.1  | 14.7  | 15.0  | 14.0  | 12.2  | 11.4  | 10.6  | (29) |  |
|                                                                        |             | MCDB   | 9.3   | 9.0   | 8.6   | 9.4   | 10.9  | 12.9  | 15.4  | 15.6  | 14.4  | 12.6  | 11.8  | 11.2  | (30) |  |
|                                                                        | 5%          | WB     | 8.3   | 8.0   | 7.8   | 8.5   | 9.7   | 11.5  | 13.9  | 14.2  | 13.2  | 11.7  | 10.8  | 9.6   | (31) |  |
|                                                                        |             | MCDB   | 8.9   | 8.7   | 8.3   | 9.1   | 10.3  | 12.2  | 14.4  | 14.8  | 13.7  | 12.3  | 11.4  | 10.2  | (32) |  |
|                                                                        | 10%         | WB     | 7.8   | 7.4   | 7.4   | 8.2   | 9.3   | 11.0  | 13.1  | 13.5  | 12.7  | 11.3  | 10.2  | 8.7   | (33) |  |
|                                                                        |             | MCDB   | 8.5   | 8.2   | 8.0   | 8.8   | 10.0  | 11.7  | 13.5  | 14.1  | 13.3  | 12.0  | 10.9  | 9.5   | (34) |  |
| (35) Mean Daily Temperature Range                                      | MDBR        |        | 2.6   | 2.7   | 2.4   | 2.4   | 2.6   | 2.4   | 2.3   | 2.2   | 2.2   | 2.2   | 2.6   | 2.8   | (35) |  |
|                                                                        | 5% DB       | MCDBR  | 2.5   | 2.7   | 2.2   | 2.4   | 2.9   | 2.9   | 3.1   | 2.9   | 2.7   | 2.1   | 2.7   | 2.9   | (36) |  |
|                                                                        |             | MCWBR  | 2.9   | 3.0   | 2.5   | 2.3   | 2.5   | 2.5   | 2.6   | 2.6   | 2.6   | 2.4   | 3.2   | 3.4   | (37) |  |
|                                                                        | 5% WB       | MCDBR  | 2.6   | 2.7   | 2.2   | 2.2   | 2.6   | 2.7   | 3.1   | 2.8   | 2.7   | 2.1   | 2.6   | 2.9   | (38) |  |
|                                                                        |             | MCWBR  | 3.1   | 3.2   | 2.6   | 2.2   | 2.4   | 2.4   | 2.7   | 2.6   | 2.7   | 2.5   | 3.1   | 3.4   | (39) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.337 | 0.356 | 0.370 | 0.387 | 0.375 | 0.369 | 0.382 | 0.369 | 0.365 | 0.349 | 0.331 | 0.422 | (40) |  |
|                                                                        | taud        |        | 2.087 | 2.201 | 2.276 | 2.288 | 2.393 | 2.450 | 2.430 | 2.468 | 2.434 | 2.321 | 2.142 | 2.429 | (41) |  |
|                                                                        | Ebn at Noon |        | 385   | 616   | 748   | 808   | 852   | 863   | 840   | 825   | 756   | 625   | 391   | 262   | (42) |  |
|                                                                        | Edh at Noon |        | 48    | 74    | 93    | 109   | 105   | 101   | 101   | 91    | 80    | 65    | 44    | 33    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 0.24  | 0.77  | 1.89  | 3.46  | 4.81  | 4.93  | 4.47  | 3.67  | 2.23  | 1.11  | 0.35  | 0.15  | (44) |  |
|                                                                        | RadStd      |        | 0.02  | 0.08  | 0.14  | 0.24  | 0.34  | 0.48  | 0.39  | 0.33  | 0.20  | 0.11  | 0.03  | 0.01  | (45) |  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (15 neighbors) |  | +0.32 | N/A     | +0.78  | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page